Year 3 CSE · Mobile & System Design

Push Notifications with Firebase Cloud Messaging

Lab 6 Flutter & FCM Integration

Lab report document.

Reg nº: 223000471

Objective

Lab Objective

Build a Flutter mobile application that integrates **Firebase Cloud Messaging (FCM)** to receive push notifications sent from the Firebase Console. The app must display the FCM device token, show received notifications in a persistent list and pop up a dialog whenever a notification arrives while the app is in the foreground.

Scope

The app was tested in two states: **foreground** (app open) and **background** (app minimised). Both scenarios were handled.

Methodology

Key Steps Followed

1 Created a new Flutter project

Ran flutter create lab6_fcm --org com.lab6 to generate the project with package ID com.lab6.lab6_fcm.

2 Created a Firebase project

Opened **console.firebase.google.com**, created a new project named *Lab6FCM*, and registered an Android app using the package name com.lab6.lab6_fcm.

3 Downloaded and placed google-services.json

Downloaded the configuration file from the Firebase Console and placed it inside android/app/ so that the Google Services Gradle plugin could find it.

4 Configured Gradle for Google Services (Kotlin DSL)

Added id("com.google.gms.google-services") version "4.4.2" apply false to settings.gradle.kts and applied it in android/app/build.gradle.kts.

5 Added Flutter dependencies

Added firebase_core, firebase_messaging, and provider to pubspec.yaml, then ran flutter pub get.

6 Added POST_NOTIFICATIONS permission

Required for Android 13 (API 33+). Added to AndroidManifest.xml so the runtime permission dialog appears on first launch.

7 Implemented FCM service & state management

Created FcmService to request permissions, retrieve the token, and attach listeners for foreground (onMessage), background tap (onMessageOpenedApp), and cold-start tap (getInitialMessage). A NotificationProvider (ChangeNotifier) stores the token and message list.

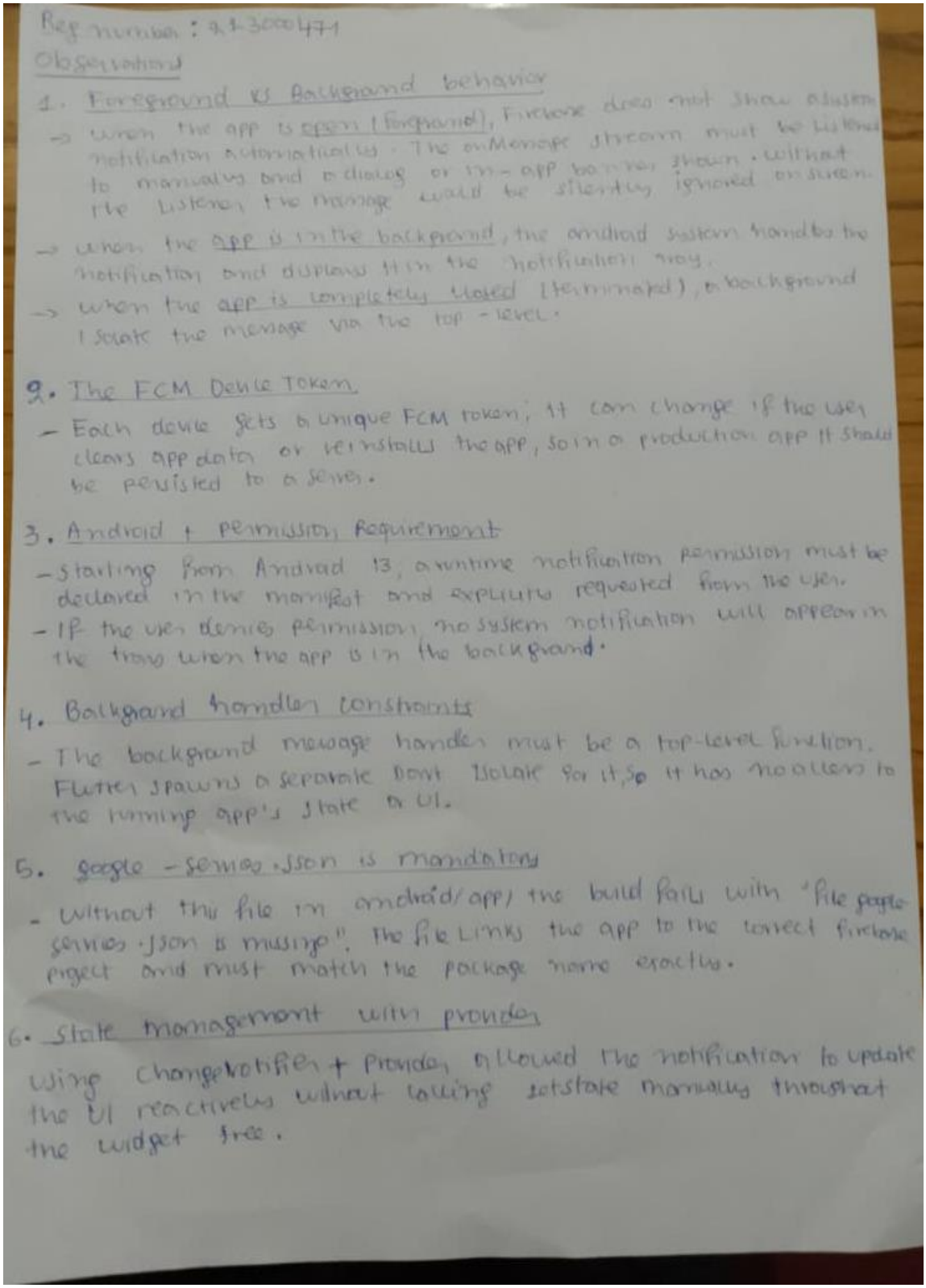
8 Built the UI

Home screen shows the FCM token in a card with a copy button, and a live list of received notifications with title, body, and timestamp. A dialog pops up automatically on foreground notifications.

9 Tested on a real Android device

Ran flutter run on a physical device, copied the token from the app, and sent test notifications from Firebase Console — both with the app open and minimised.

Analysis and Observations



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Observations

1. Foreground vs Background behavior

→ when the app is open (foreground), Firebase does not show a system notification automatically. The onMessage stream must be listened to manually and a dialog or in-app banner shown. without the listener, the message would be silently ignored on screen.

→ when the app is in the background, the android system handles the notification and displays it in the notification tray.

→ when the app is completely closed (terminated), a background isolate the message via the top-level.

2. The FCM Device Token

- Each device gets a unique FCM token; it can change if the user clears app data or reinstalls the app, so in a production app it should be persisted to a server.

3. Android 13 permission Requirement

- Starting from Android 13, a runtime notification permission must be declared in the manifest and explicitly requested from the user.
- If the user denies permission, no system notification will appear in the tray when the app is in the background.

4. Background handler constraints

- The background message handler must be a top-level function. Flutter spawns a separate Dart isolate for it, so it has no access to the running app's state or UI.

5. google-services.json is mandatory

- without this file in android/app/ the build fails with "File google-services.json is missing". The file links the app to the correct Firebase project and must match the package name exactly.

6. State management with provider

Using ChangeNotifier + Provider allowed the notification to update the UI reactively without calling setstate manually throughout the widget tree.

Figure 1 Observations

Screenshot

App UI — Home Screen

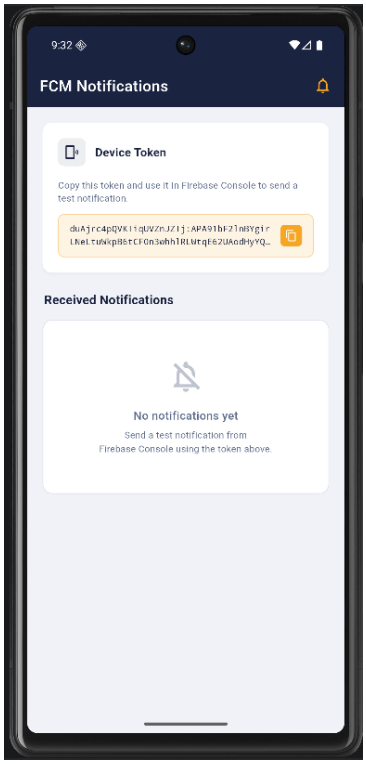


Figure 2 Home Screen — Token Displayed

Foreground Notification Popup

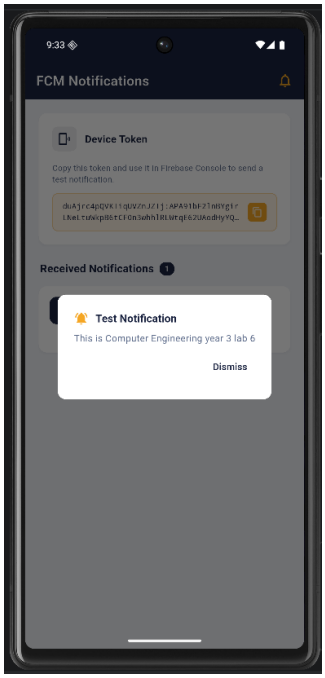


Figure 3 Notification popup

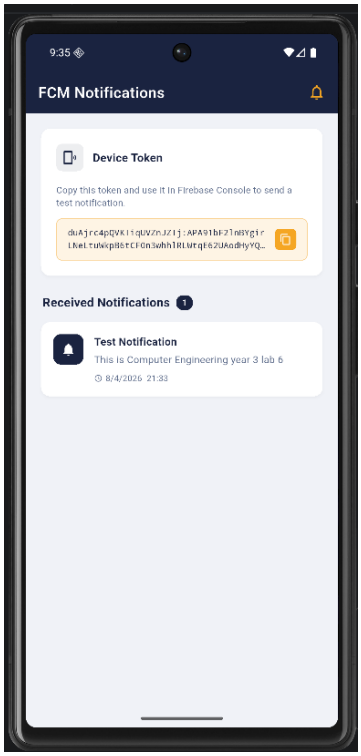


Figure 4 Received Notification lists

Background Notification

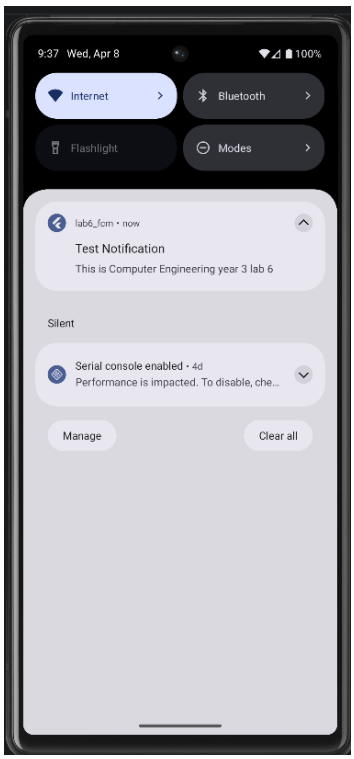


Figure 5 Background Notification

Firebase Console Screenshots

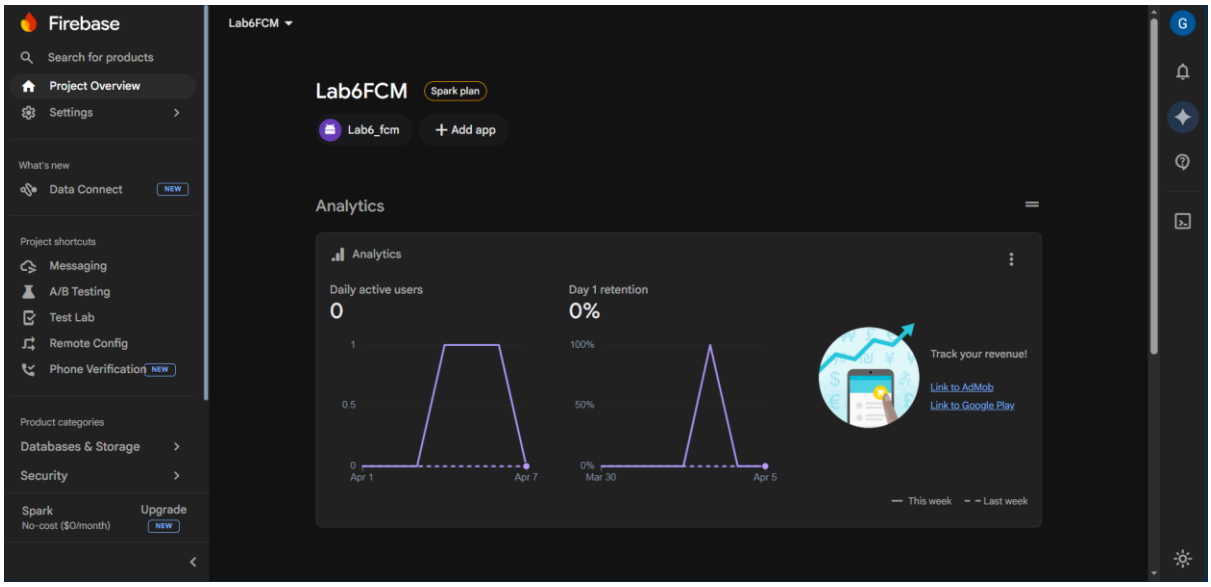


Figure 6Firebase Project Overview

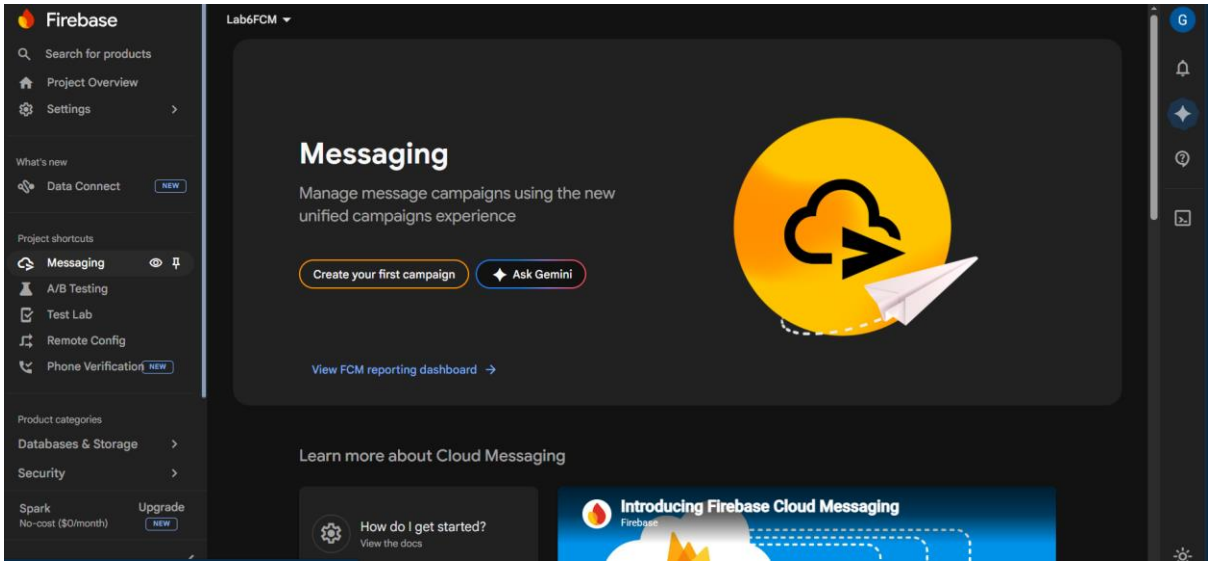


Figure 7 Cloud Messaging — New Message

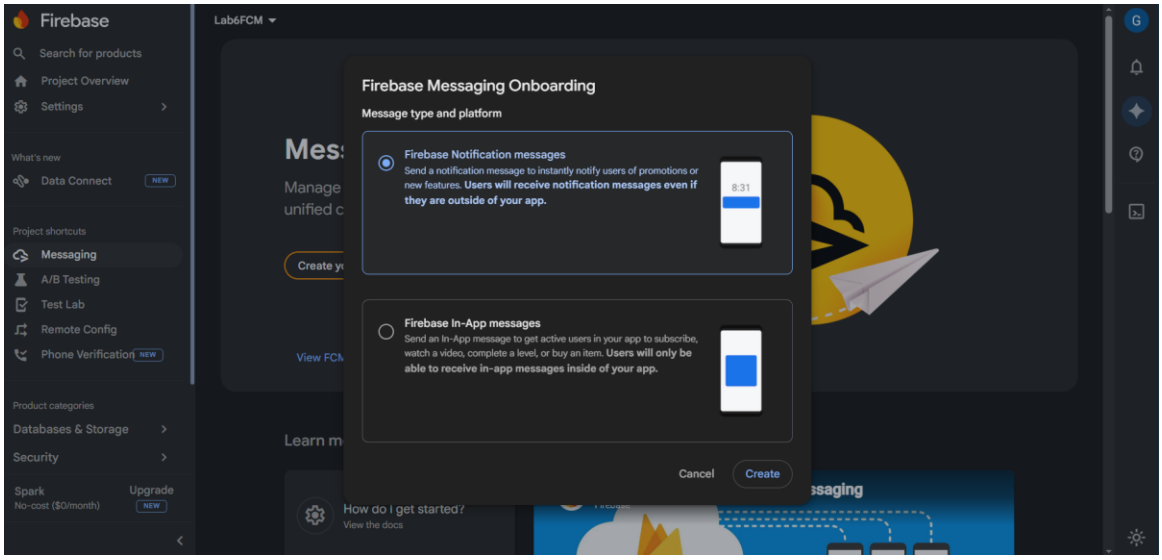
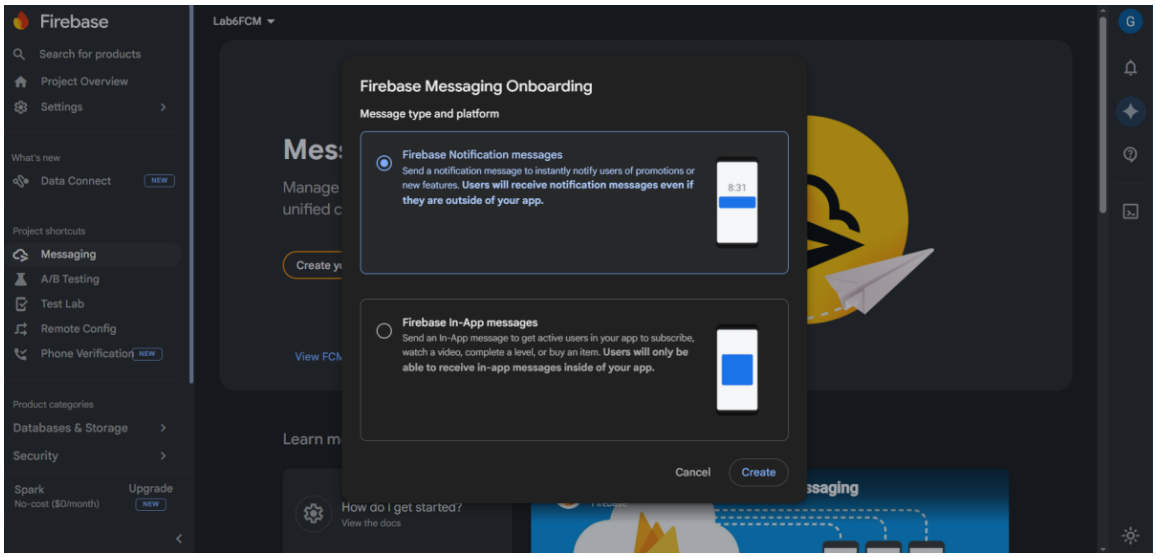


Figure 8 Firebase Notification Message



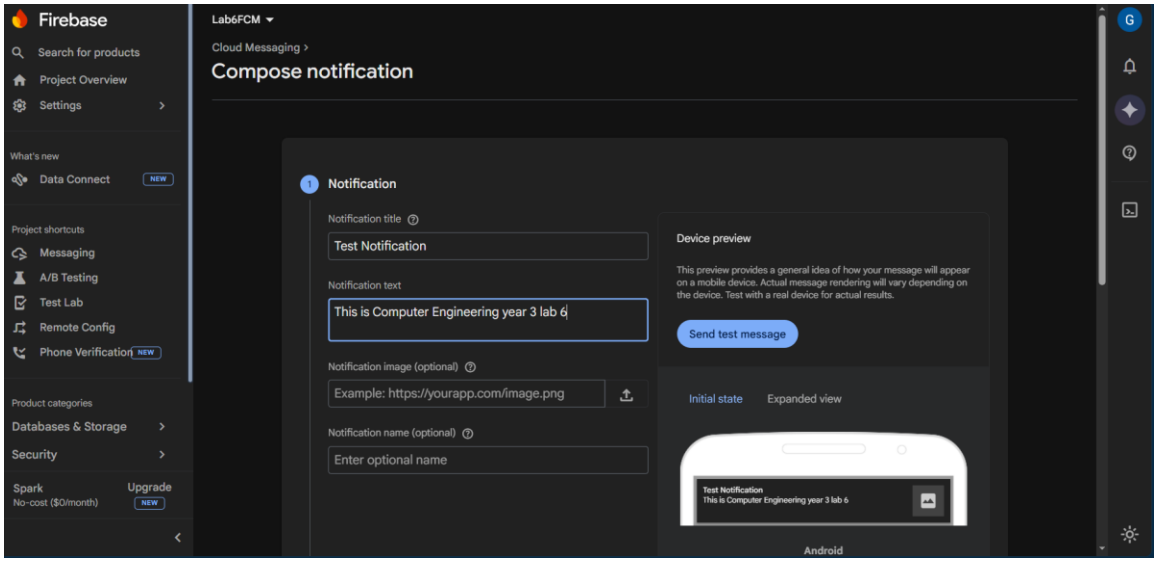


Figure 9 Test notification

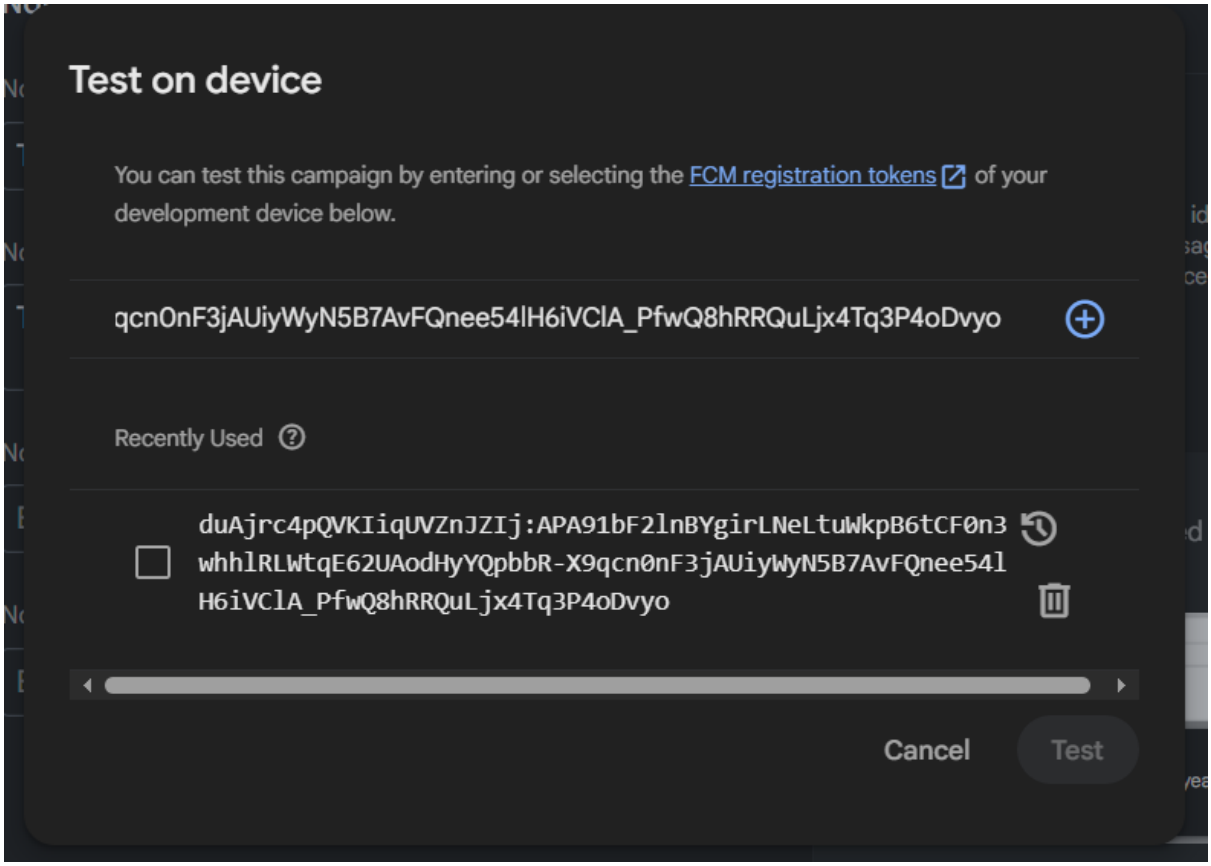


Figure 10 Adding device notification